

PRAKTIKUM SISTEM OPERASI

MODUL V

EKSPLORASI PROSES XINU



Oleh:

Marvel Sanjaya Setiawan

2311104053

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

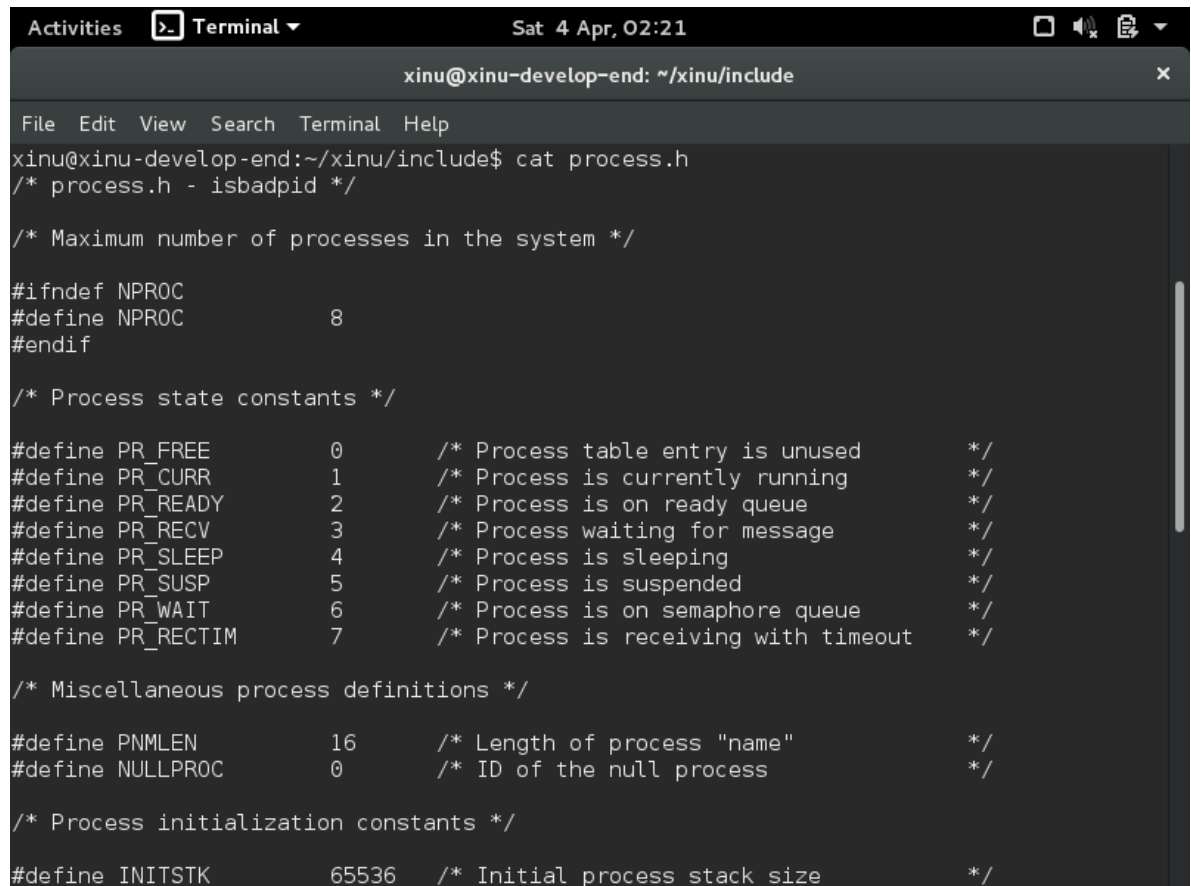
2026

JURNAL

1. Jawablah pertanyaan berikut ini:

- a. Berapa banyaknya maksimum proses yang ada pada Xinu?**
- b. Berapa maksimal panjang nama suatu proses pada Xinu?**
- c. Berapa nilai prioritas awal pada saat proses dibuat?**
- d. Ada berapa total state pada Xinu? Sebutkan!**

Jawaban:



```
Activities Terminal Sat 4 Apr, 02:21
xinu@xinu-develop-end: ~/xinu/include

File Edit View Search Terminal Help
xinu@xinu-develop-end:~/xinu/include$ cat process.h
/* process.h - isbadpid */

/* Maximum number of processes in the system */

#ifndef NPROC
#define NPROC      8
#endif

/* Process state constants */

#define PR_FREE      0      /* Process table entry is unused */
#define PR_CURR      1      /* Process is currently running */
#define PR_READY     2      /* Process is on ready queue */
#define PR_RECV      3      /* Process waiting for message */
#define PR_SLEEP     4      /* Process is sleeping */
#define PR_SUSP      5      /* Process is suspended */
#define PR_WAIT      6      /* Process is on semaphore queue */
#define PR_RECTIM    7      /* Process is receiving with timeout */

/* Miscellaneous process definitions */

#define PNMLEN        16     /* Length of process "name" */
#define NULLPROC      0      /* ID of the null process */

/* Process initialization constants */

#define INITSTK        65536 /* Initial process stack size
```

```
xinu@xinu-develop-end: ~/xinu/include
File Edit View Search Terminal Help
#define INITSTK          65536 /* Initial process stack size */
#define INITPRIO         20  /* Initial process priority */
#define INITRET          userret /* Address to which process returns */

/* Inline code to check process ID (assumes interrupts are disabled) */

#define isbadpid(x)      ( ((pid32)(x) < 0) || \
                          ((pid32)(x) >= NPROC) || \
                          (proctab[(x)].prstate == PR_FREE))

/* Number of device descriptors a process can have open */

#define NDESC            5    /* Must be odd to make procent 4N bytes */

/* Definition of the process table (multiple of 32 bits) */

struct procent {          /* Entry in the process table */
    uint16 prstate;        /* Process state: PR_CURR, etc. */
    pri16  prprio;         /* Process priority */
    char *prstkptr;        /* Saved stack pointer */
    char *prstkbase;       /* Base of run time stack */
    uint32 prstklen;       /* Stack length in bytes */
    char prname[PNMLEN];   /* Process name */
    sid32  prsem;          /* Semaphore on which process waits */
    pid32  prparent;       /* ID of the creating process */
    umsg32 prmsg;          /* Message sent to this process */
    bool8  prhasmsg;       /* Nonzero iff msg is valid */
    int16  prdesc[NDESC]; /* Device descriptors for process */
};
```

```
Activities Terminal Sat 4 Apr, 02:23 xinu@xinu-develop-end: ~/xinu/include
File Edit View Search Terminal Help
        (proctab[(x)].prstate == PR_FREE))

/* Number of device descriptors a process can have open */
#define NDESC          5          /* Must be odd to make procent 4N bytes */

/* Definition of the process table (multiple of 32 bits) */
struct procent {
    uint16 prstate;          /* Entry in the process table */
    pri16 prprio;            /* Process state: PR_CURR, etc. */
    char *prstkptr;          /* Process priority */
    char *prstkbase;         /* Saved stack pointer */
    uint32 prstklen;         /* Base of run time stack */
    char prname[PNMLEN];     /* Stack length in bytes */
    sid32 prsem;             /* Process name */
    pid32 prparent;          /* Semaphore on which process waits */
    umsg32 prmsg;            /* ID of the creating process */
    bool8 prhasmsg;          /* Message sent to this process */
    int16 prdesc[NDESC];     /* Nonzero iff msg is valid */
};                             /* Device descriptors for process */

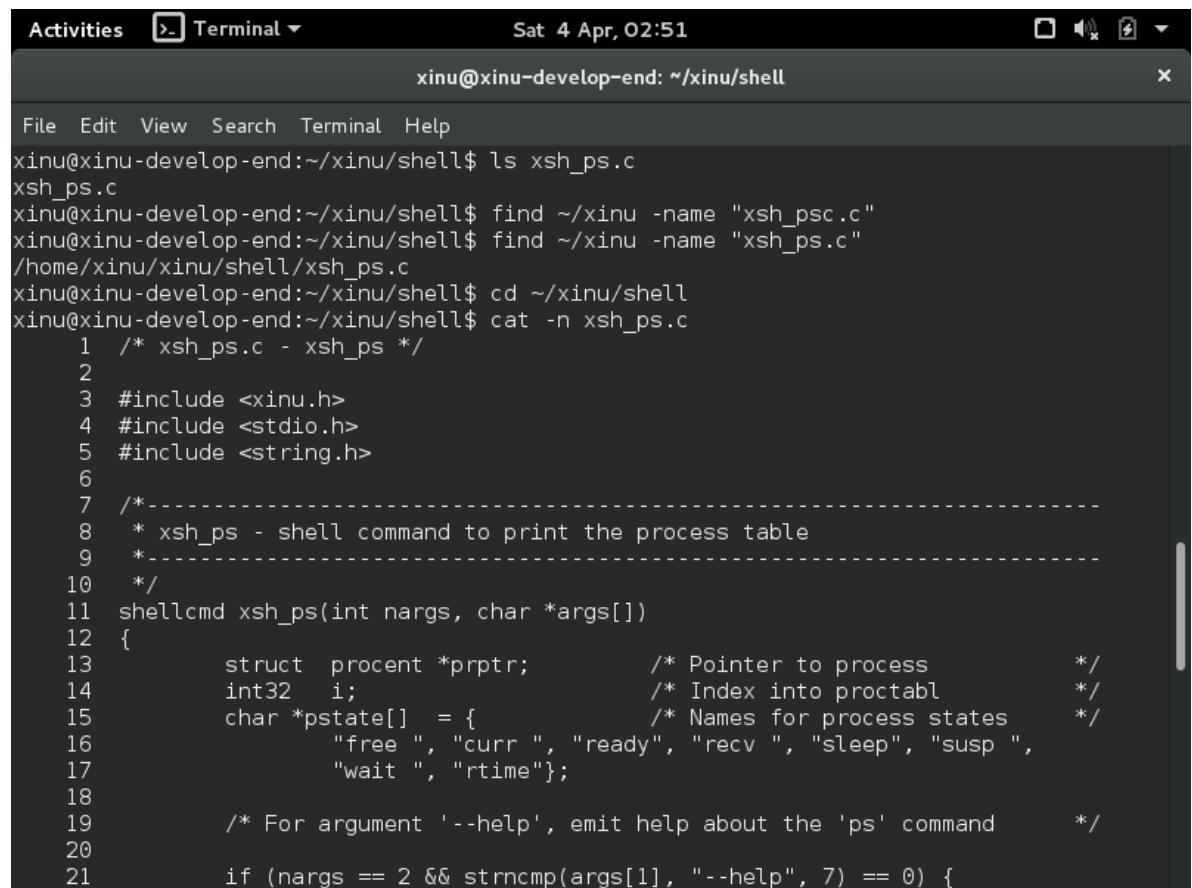
/* Marker for the top of a process stack (used to help detect overflow) */
#define STACKMAGIC      0x0A0AAAA9

extern struct procent proctab[];
extern int32 prcount;        /* Currently active processes */
extern pid32 currpri;        /* Currently executing process */
```

- a. 8 proses
- b. 16 karakter
- c. 20

2. Komentar xsh_ps.c

Jawaban:



```
Activities Terminal Sat 4 Apr, 02:51
xinu@xinu-develop-end: ~/xinu/shell

File Edit View Search Terminal Help
xinu@xinu-develop-end:~/xinu/shell$ ls xsh_ps.c
xsh_ps.c
xinu@xinu-develop-end:~/xinu/shell$ find ~/xinu -name "xsh_psc.c"
xinu@xinu-develop-end:~/xinu/shell$ find ~/xinu -name "xsh_ps.c"
/home/xinu/xinu/shell/xsh_ps.c
xinu@xinu-develop-end:~/xinu/shell$ cd ~/xinu/shell
xinu@xinu-develop-end:~/xinu/shell$ cat -n xsh_ps.c
 1  /* xsh_ps.c - xsh_ps */
 2
 3  #include <xinu.h>
 4  #include <stdio.h>
 5  #include <string.h>
 6
 7  /*-----
 8   * xsh_ps - shell command to print the process table
 9   *-----
10   */
11  shellcmd xsh_ps(int nargs, char *args[])
12  {
13      struct procent *prptr;          /* Pointer to process */
14      int32 i;                        /* Index into proctabl */
15      char *pstate[] = {              /* Names for process states */
16          "free ", "curr ", "ready", "recv ", "sleep", "susp ",
17          "wait ", "rtime"};
18
19      /* For argument '--help', emit help about the 'ps' command */
20
21      if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
```

```
Activities Terminal Sat 4 Apr, 02:48
xinu@xinu-develop-end: ~/xinu/shell
File Edit View Search Terminal Help
GNU nano 2.2.6 File: xsh_ps.c Modified

/* xsh_ps.c - xsh_ps */
/* Nama:[Marvel] - Nim:[2311104053] */

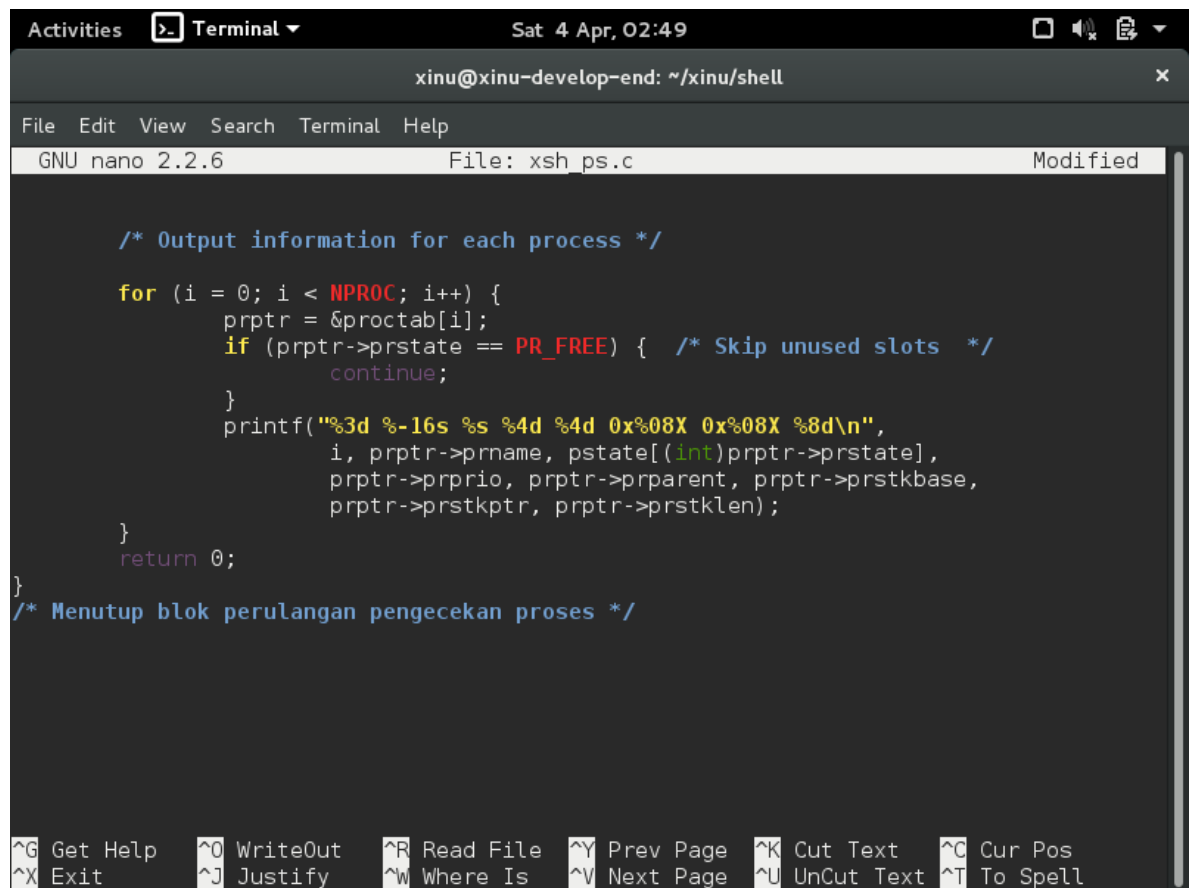
#include <xinu.h>
#include <stdio.h>
#include <string.h>

/*-----
 * xsh_ps - shell command to print the process table
 *-----
 */
shellcmd xsh_ps(int nargs, char *args[])
{
    struct proctent *prptr;          /* Pointer to process */
    int32 i;                          /* Index into proctabl */
    char *pstate[] = {               /* Names for process states */
        "free ", "curr ", "ready", "recv ", "sleep", "susp ",
        "wait ", "rtime"};

    /* For argument '--help', emit help about the 'ps' command */

    if (nargs == 2 && strncmp(args[1], "--help", 7) == 0) {
        printf("Use: %s\n\n", args[0]);
    }
}

^G Get Help    ^O WriteOut    ^R Read File   ^Y Prev Page   ^K Cut Text     ^C Cur Pos
^X Exit        ^J Justify     ^W Where Is    ^V Next Page   ^U UnCut Text  ^T To Spell
```



```
/* Output information for each process */

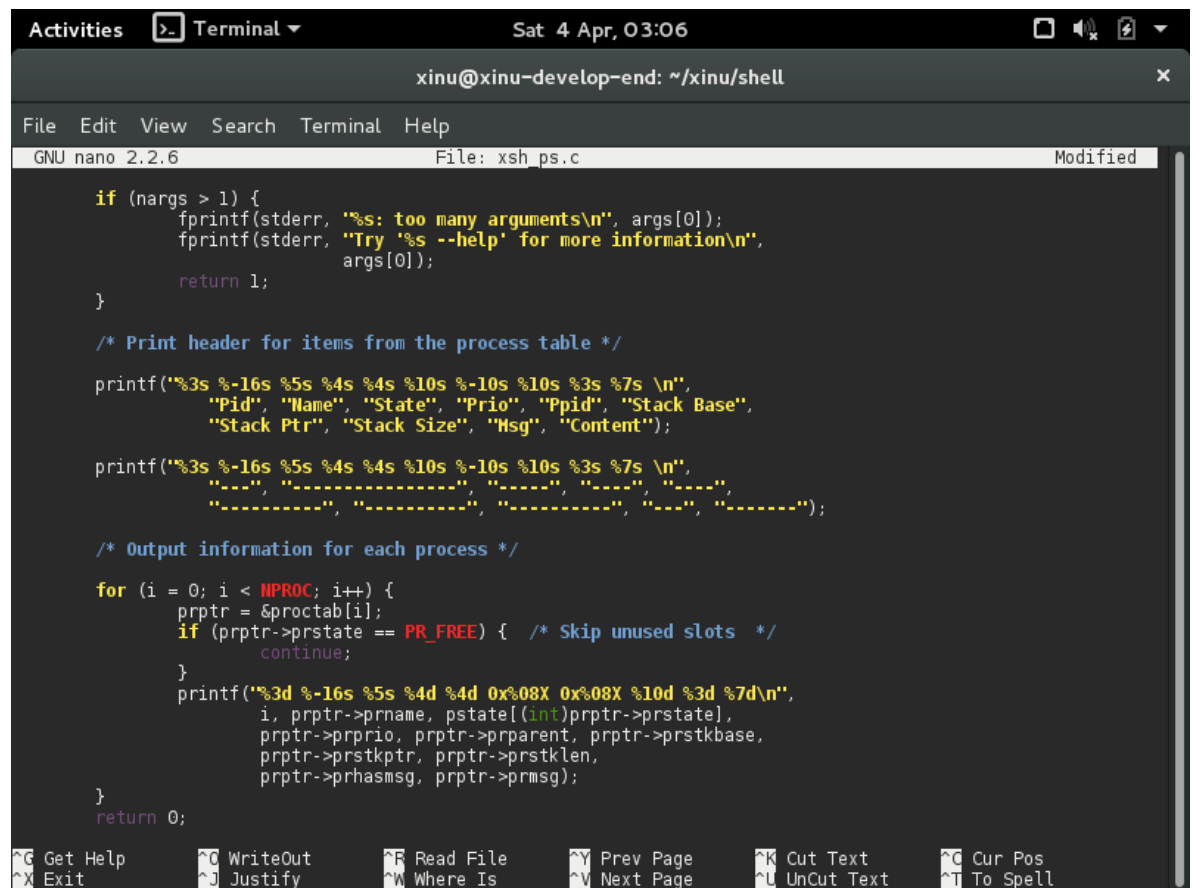
for (i = 0; i < NPROC; i++) {
    prptr = &proctab[i];
    if (prptr->prstate == PR_FREE) { /* Skip unused slots */
        continue;
    }
    printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
        i, prptr->prname, pstate[(int)prptr->prstate],
        prptr->prprio, prptr->prparent, prptr->prstkbase,
        prptr->prstkptr, prptr->prstklen);
}
return 0;
}
/* Menutup blok perulangan pengecekan proses */
```

Urutan pengerjaan:

1. Masuk ke folder shell → `cd ~/xinu/shell`
2. Lihat isi file → `cat -n xsh_ps.c`
3. Edit file → `nano xsh_ps.c`
4. Tambahkan komentar `/* ... */` di 20 baris terakhir (baris 43-62)
5. Tambahkan identitas nama/NIM di baris paling atas
6. Save → `Ctrl+X` → `Y` → `Enter`

3. Tambah Kolom Msg & Content di perintah ps

Jawaban:



```
Activities Terminal Sat 4 Apr, 03:06
xinu@xinu-develop-end: ~/xinu/shell
File Edit View Search Terminal Help
GNU nano 2.2.6 File: xsh_ps.c Modified

    if (nargs > 1) {
        fprintf(stderr, "%s: too many arguments\n", args[0]);
        fprintf(stderr, "Try '%s --help' for more information\n",
            args[0]);
        return 1;
    }

    /* Print header for items from the process table */

    printf("%3s %-16s %5s %4s %4s %10s %-10s %10s %3s %7s \n",
        "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
        "Stack Ptr", "Stack Size", "Msg", "Content");

    printf("%3s %-16s %5s %4s %4s %10s %-10s %10s %3s %7s \n",
        "----", "-----", "-----", "-----", "-----",
        "-----", "-----", "-----", "----", "-----");

    /* Output information for each process */

    for (i = 0; i < NPROC; i++) {
        prptr = &proctab[i];
        if (prptr->prstate == PR_FREE) { /* Skip unused slots */
            continue;
        }
        printf("%3d %-16s %5s %4d %4d 0x%08X 0x%08X %10d %3d %7d\n",
            i, prptr->prname, prptr->prstate,
            prptr->prprio, prptr->prparent, prptr->prstkbase,
            prptr->prstkptr, prptr->prstklen,
            prptr->prhasmsg, prptr->prmsg);
    }

    return 0;

^G Get Help      ^O WriteOut      ^R Read File     ^Y Prev Page     ^K Cut Text      ^C Cur Pos
^X Exit          ^J Justify       ^W Where Is      ^V Next Page     ^L UnCut Text    ^T To Spell
```

```
Activities Terminal Sat 4 Apr, 04:22
xinu@xinu-develop-end: ~
File Edit View Search Terminal Help

-----MARVEL-----
XINU
-----2311104053-----

Welcome to Xinu!
Nama: [Marvel]
NIM: [2311104053]

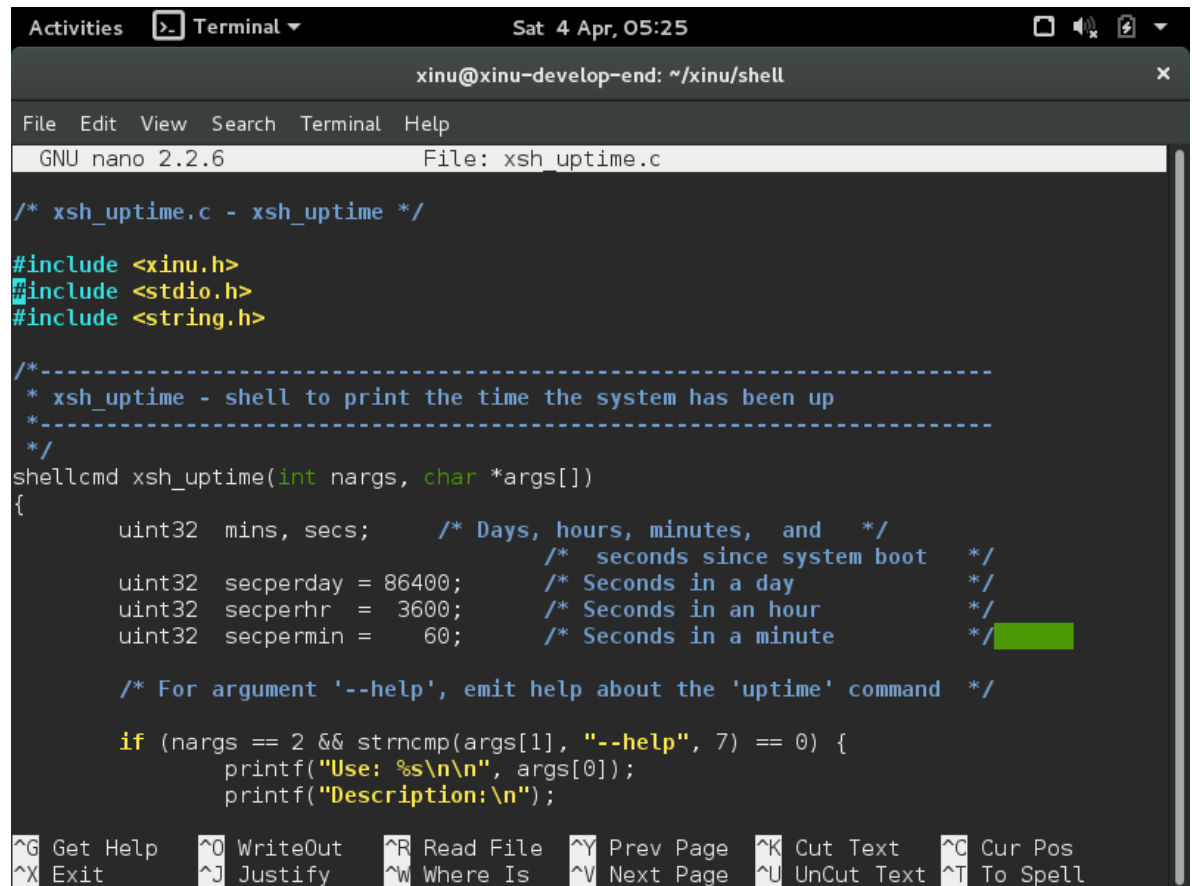
xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFEE4    8192    1    3
1 ipout           wait   500  0 0x005FBFFC 0x005FBF30    8192    0    0
2 netin           wait   500  0 0x005F9FFC 0x005F9ED0    8192    0    0
4 Main process    recv   20    3 0x005E7FFC 0x005E7F34   65536    0    0
5 shell           recv   50    4 0x005D7FFC 0x005D79AC    8192    0    0
6 ps              curr   20    5 0x005F7FFC 0x005F7FC4    8192    0    0
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

Urutan pengerjaan:

1. Edit file → nano xsh_ps.c
2. Ubah header printf → tambahkan "Msg" dan "Content"
3. Ubah separator printf → tambahkan "---" dan "-----"
4. Ubah data printf → tambahkan prptr->prhasmsg dan prptr->prmsg
5. Save → Ctrl+X → Y → Enter
6. Compile → cd ~/xinu/compile && make clean && make
7. Jalankan backend VM → sudo minicom
8. Ketik ps di Xinu shell
9. Screenshot source code + output

4. Uptime dalam menit

Jawaban:



```
Activities Terminal Sat 4 Apr, 05:25
xinu@xinu-develop-end: ~/xinu/shell
File Edit View Search Terminal Help
GNU nano 2.2.6 File: xsh_uptime.c

/* xsh_uptime.c - xsh_uptime */

#include <xinu.h>
#include <stdio.h>
#include <string.h>

/*-----
 * xsh_uptime - shell to print the time the system has been up
 *-----
 */
shellcmd xsh_uptime(int nargs, char *args[])
{
    uint32 mins, secs;      /* Days, hours, minutes, and */
                           /* seconds since system boot */
    uint32 secperday = 86400; /* Seconds in a day */
    uint32 secperhr = 3600;  /* Seconds in an hour */
    uint32 secpermin = 60;   /* Seconds in a minute */

    /* For argument '--help', emit help about the 'uptime' command */

    if (nargs == 2 && strncmp(args[1], "--help", 7) == 0) {
        printf("Use: %s\n\n", args[0]);
        printf("Description:\n");
    }
}
```

Activities Terminal Sat 4 Apr, 05:25

xinu@xinu-develop-end: ~/xinu/shell

File Edit View Search Terminal Help

GNU nano 2.2.6 File: xsh_uptime.c

```
}

secs = clktime;      /* Total seconds since boot */

/* Subtract number of minutes */

mins = secs/secpermin;

printf("Xinu has been up %d minute(s)\n", mins);

return 0;
}
```

^G Get Help ^O WriteOut ^R Read File ^Y Prev Page ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell

```
Activities Terminal Sat 4 Apr, 05:27
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

[0x00100000 to 0x0011C2DE]
148488 bytes of data.
[0x0011FEC0 to 0x001442C7]

Obtained IP address 10.0.3.15 (0x0a00030f)

-----MARVEL-----

XINU

-----2311104053-----

Welcome to Xinu!
Nama: [Marvel]
NIM: [2311104053]

xsh $ uptime
Xinu has been up 0 minute(s)
xsh $

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

Urutan pengerjaan:

1. Edit file → nano ~/xinu/shell/xsh_uptime.c
2. Ubah variabel baris 13 → hapus days, hrs, sisakan mins, secs
3. Hapus semua perhitungan days/hours/seconds
4. Ganti output dengan → printf("Xinu has been up %d minute(s)\n", mins)
5. Save → Ctrl+X → Y → Enter
6. Compile → cd ~/xinu/compile && make clean && make
7. Jalankan backend VM → sudo minicom
8. Ketik uptime di Xinu shell
9. Screenshot source code + output